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This project investigated how chemical and oxidative stressors, in combination with desiccation duration, influence the growth dynamics of Group B Streptococcus (GBS). Specifically, the study examined three stress conditions ethanol, bleach, and paraquat (oxidative stress) applied to multiple GBS strains (A909 and derivatives), with cells previously subjected to different desiccation periods (including 74 and 105 days). The scientific question focused on whether stress exposure and desiccation history alter bacterial growth performance, as measured by optical density (OD600) over time.

The dataset originated as a raw plate reader export file containing time-series absorbance measurements across multiple wells. Significant preprocessing was required to prepare the data for analysis. Using R, the relevant measurement region was extracted from the raw file, column headers were reassigned based on plate metadata, and all OD values were converted to numeric format. The dataset was then reshaped from wide to long format using pivot_longer(), enabling consistent handling of time-series measurements across all wells. A key step involved constructing experimental metadata that mapped each well to its corresponding condition (strain, treatment concentration, and desiccation duration). Time values were converted into hours using hms, and observations with missing or incomplete timestamps were filtered out to ensure data integrity.

For analysis, the data were grouped by time and experimental condition, and summary statistics were calculated, including mean OD, standard deviation, and standard error. These summaries allowed for comparison of growth trajectories across conditions while accounting for replicate variability. Three major treatment-specific analyses were conducted: bleach exposure, ethanol exposure, and paraquat exposure. For each condition, growth curves were generated using ggplot2, with mean OD plotted over time and variability displayed using standard error ribbons.

The results show that all stressors reduced bacterial growth relative to less stressful conditions, though the magnitude and timing of inhibition varied by treatment. Bleach exposure produced the strongest overall suppression of growth, while ethanol showed concentration-dependent effects. Paraquat exposure resulted in more variable growth patterns, suggesting strain-specific or dose-dependent oxidative stress responses. Additionally, differences between desiccation timepoints (74 vs. 105 days) suggest that prolonged desiccation may influence stress tolerance, although effects were not consistent across all treatments.

Overall, these findings indicate that GBS growth is shaped by both environmental stress exposure and prior desiccation history. However, interpretation is limited by biological variability among replicates and the absence of formal statistical testing. Future analyses incorporating inferential statistics and expanded replicate design would strengthen conclusions regarding stress adaptation mechanisms in GBS.